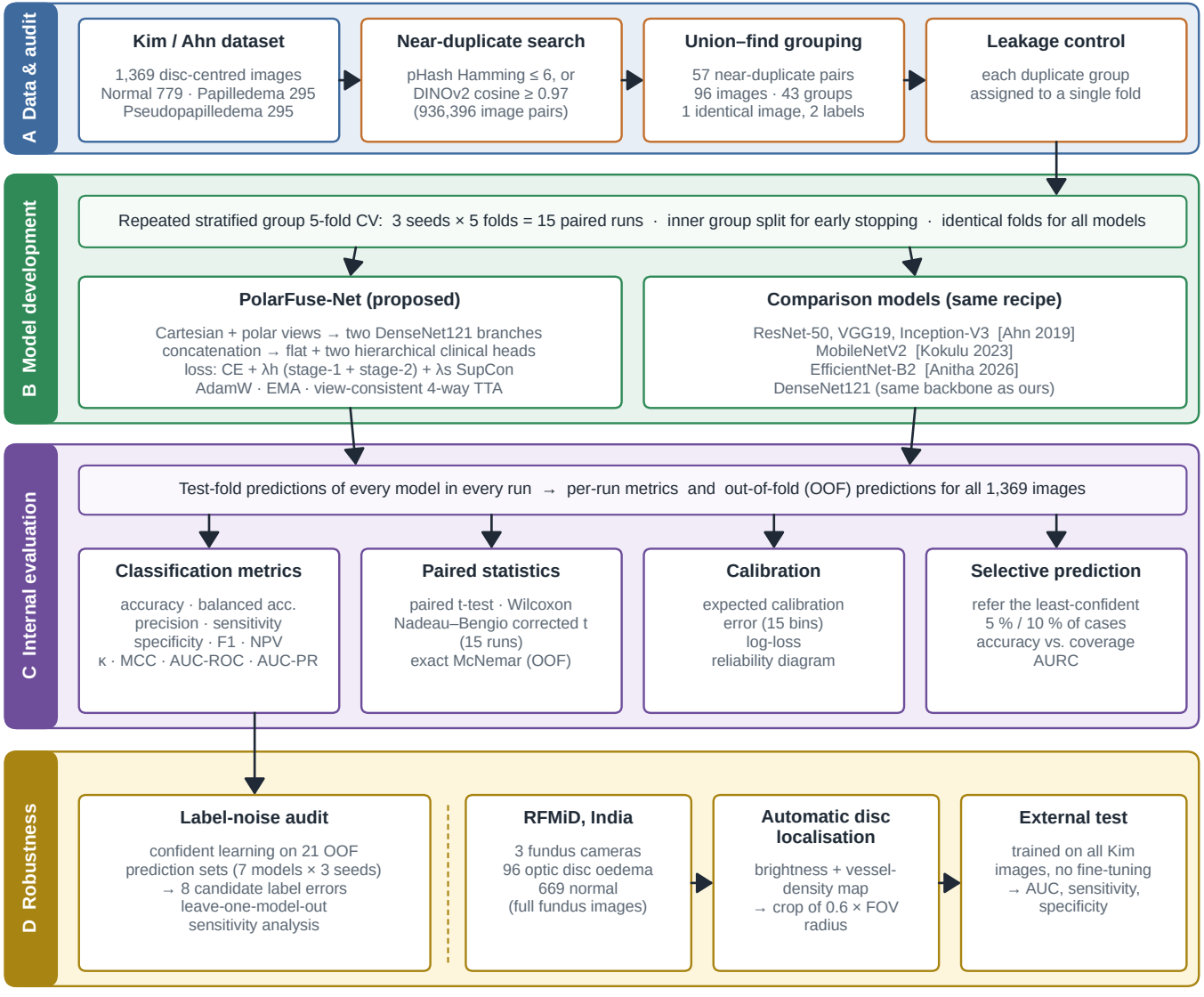
A Data & audit
Kim / Ahn dataset
1,369 disc-centred images
Normal 779 · Papilledema 295
Pseudopapilledema 295
Near-duplicate search
pHash Hamming ≤ 6, or
DINOv2 cosine ≥ 0.97
(936,396 image pairs)
Union–find grouping
57 near-duplicate pairs
96 images · 43 groups
1 identical image, 2 labels
Leakage control
each duplicate group
assigned to a single fold
B Model development
Repeated stratified group 5-fold CV: 3 seeds × 5 folds = 15 paired runs · inner group split for early stopping · identical folds for all models
PolarFuse-Net (proposed)
Cartesian + polar views → two DenseNet121 branches
concatenation → flat + two hierarchical clinical heads
loss: CE + λh (stage-1 + stage-2) + λs SupCon
AdamW · EMA · view-consistent 4-way TTA
Comparison models (same recipe)
ResNet-50, VGG19, Inception-V3 [Ahn 2019]
MobileNetV2 [Kokulu 2023]
EfficientNet-B2 [Anitha 2026]
DenseNet121 (same backbone as ours)
C Internal evaluation
Test-fold predictions of every model in every run → per-run metrics and out-of-fold (OOF) predictions for all 1,369 images
Classification metrics
accuracy · balanced acc.
precision · sensitivity
specificity · F1 · NPV
κ · MCC · AUC-ROC · AUC-PR
Paired statistics
paired t-test · Wilcoxon
Nadeau–Bengio corrected t
(15 runs)
exact McNemar (OOF)
Calibration
expected calibration
error (15 bins)
log-loss
reliability diagram
Selective prediction
refer the least-confident
5 % / 10 % of cases
accuracy vs. coverage
AURC
D Robustness
Label-noise audit
confident learning on 21 OOF
prediction sets (7 models × 3 seeds)
→ 8 candidate label errors
leave-one-model-out
sensitivity analysis
RFMiD, India
3 fundus cameras
96 optic disc oedema
669 normal
(full fundus images)
Automatic disc localisation
brightness + vessel-density map
→ crop of 0.6 × FOV radius
External test
trained on all Kim images, no fine-tuning
→ AUC, sensitivity, specificity